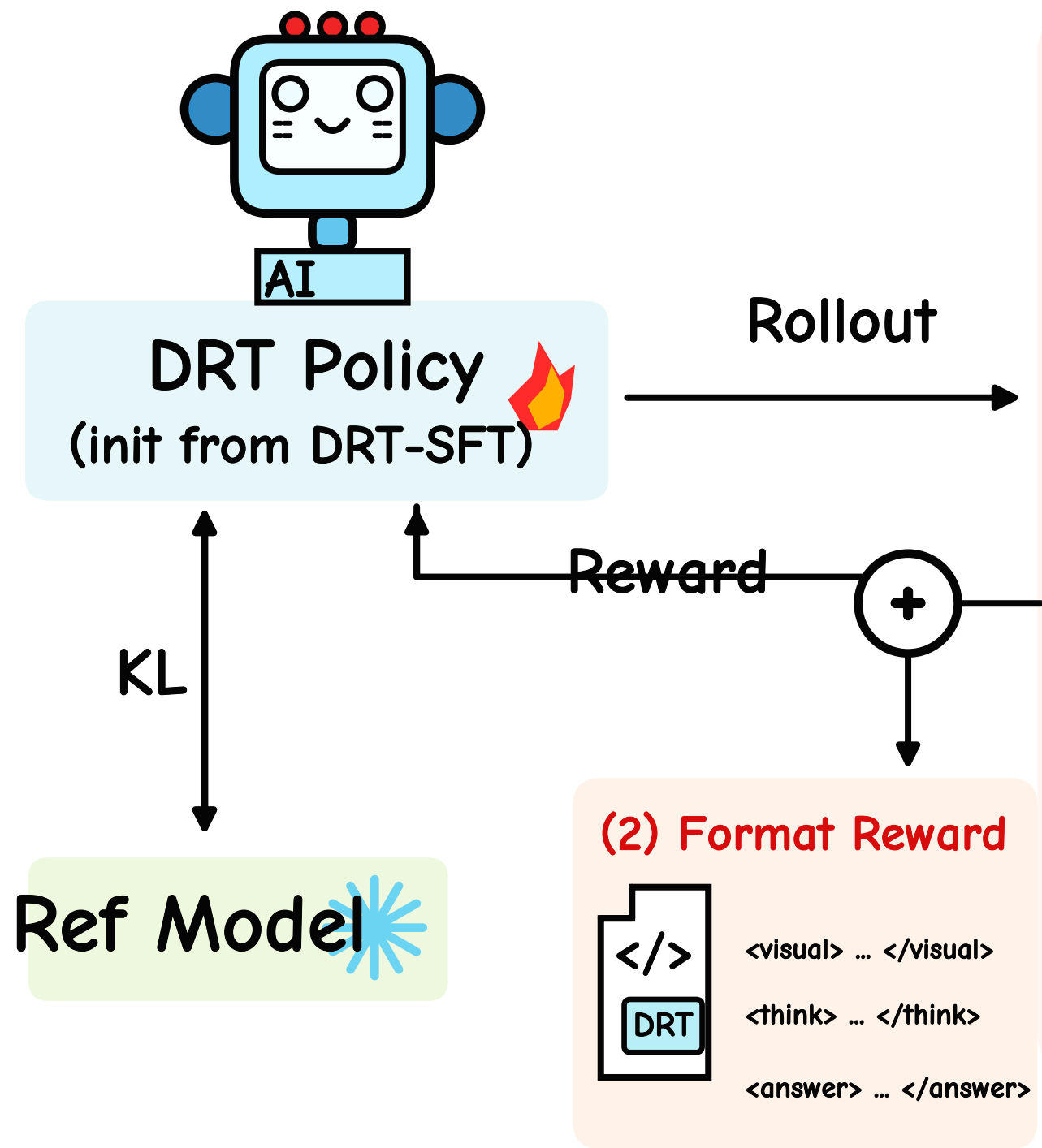




(1) Main Reward + Step Bonus



Process Reward (R_{main})

✓ → ✓ → ✓ → ✓ → ✓

1.0

✓ → ✓ → ✓ → ✓ → ✗

0.9

✓ → ! → ! → ! → !

0.2

✓ matched step ! hallucinated step ✗ mismatch step

Step Bonus

effective multi-step

● → ● → ● → ● → ●

+
bonus

deep exploration

● → ● → ● → ●
└───┘
└───┘

++
bonus